

Artemy Kolchinsky

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Last updated: July 15, 2026

RESEARCH PROFILE I am a theorist working at the interface of nonequilibrium statistical physics and information theory. My work focuses on fundamental bounds, tradeoffs, and decompositions that govern information processing in nonequilibrium systems, ranging from molecular replicators to neural networks.

RESEARCH POSITIONS

2023–	Universitat Pompeu Fabra (Barcelona, Spain). Marie Curie Fellow.	Host: Ricard Solé
2022–2023	University of Tokyo, Universal Biology Institute. Project Researcher.	PI: Sosuke Ito
2015–2021	Santa Fe Institute (Santa Fe, USA). Program Postdoctoral Fellow.	PI: David H. Wolpert
2014	LinkedIn Corporation, Data Science Research Intern.	
2009–2010	Instituto Gulbenkian de Ciência (Oeiras, Portugal). Predoctoral Visiting Researcher.	

AFFILIATIONS

2024–	Barcelona Collaboratorium for Modelling and Predictive Biology. Associated fellow.
2023–	University of Tokyo, Universal Biology Institute. Affiliated researcher.

EDUCATION

2015	Indiana University. Ph.D. in Informatics (Complex Systems focus), Minor in Cognitive Science Thesis: “Measuring scales: integration and modularity in complex systems”. Advisor: Luis M. Rocha
2004	New York University, B.A. <i>Magna Cum Laude</i>

PUBLICATIONS *Equal contribution †Corresponding author

A. Kolchinsky, “Elementary derivation of the dissipation–coherence bound for stochastic oscillators”, *Physical Review E*, in press. [arXiv:2510.14101 link](#)

J. Bauermann, G. Bartolucci, **A. Kolchinsky**, “Spatiotemporal organization of chemical oscillators via phase separation”, *Physical Review Letters*, 136, 178001 (2026). [pdf link](#)

A. Kolchinsky[†], A. Dechant, K. Yoshimura, S. Ito, “Generalized free energy and excess/housekeeping decomposition in nonequilibrium systems: from large deviations to thermodynamic speed limits”, *Physical Review Research*, 8, 023025 (2026). [pdf link](#)

J. Piñero, D. R. Sowinski, G. Ghoshal, A. Frank, **A. Kolchinsky**[†], “Information bounds production in replicator systems”, *Communications Physics*, 9, 120 (2026). [pdf link](#)

M. Aguilera^{*†}, S. Ito, **A. Kolchinsky**^{*†}, “Inferring entropy production in many-body systems using nonequilibrium maximum entropy”, *Physical Review Letters*, 136, 077101 (2026). [pdf link](#)

OoLEN collaboration, “What it takes to solve the Origin(s) of Life: An integrated review. Part 1: Experimental methods and data repositories”, *Cell Reports Physical Science* (2026). [pdf link](#)

OoLEN collaboration, “What it takes to solve the Origin(s) of Life: An integrated review. Part 2: Theoretical methods and emerging trends”, *Cell Reports Physical Science* (2026). [pdf link](#)

G.-H. Xu, **A. Kolchinsky**, J.-C. Delvenne, S. Ito, “Thermodynamic geometric constraint on the spectrum of Markov rate matrices”, *Physical Review Letters*, 135, 257102 (2025). [pdf link](#) (*Editors’ Suggestion*)

J. Piñero^{*}, **A. Kolchinsky**^{*}, S. Redner, R. Solé, “Neutral theory of cooperative dynamics”, *Proceedings of the National Academy of Sciences*, 122, 51 (2025). [pdf link](#)

A. Kolchinsky, “Thermodynamics of Darwinian evolution in molecular replicators”, *Philosophical Transactions of the Royal Society B* 380, 20240436 (2025). [pdf link](#)

K. R. Thomsen, **A. Kolchinsky**, S. Rasmussen, “Protocellular energy transduction, information, and fitness”, *Philosophical Transactions of the Royal Society B* 380, 20240294 (2025). [pdf link](#)

S. Bartlett, A. W. Eckford, M. Egbert, M. Lingam, **A. Kolchinsky**[†], A. Frank[†], G. Ghoshal[†], “The Physics of Life: Exploring Information as a Distinctive Feature of Living Systems”, *PRX Life* 3, 037003 (2025). [pdf link](#)

R. Nagayama, K. Yoshimura, **A. Kolchinsky**, S. Ito, “Geometric thermodynamics of reaction-diffusion systems: Thermodynamic trade-off relations and optimal transport for pattern formation”, *Physical Review Research* 7, 033011 (2025). [pdf link](#)

A. Kolchinsky[†], I. Marvian, C. Gokler, Z. Liu, P. Shor, O. Shtanko, K. Thompson, D. Wolpert, S. Lloyd, “Maximizing free energy gain”, *Entropy* 27, 91 (2025). [pdf link](#)

A. Kolchinsky, “Thermodynamic dissipation does not bound replicator growth and decay rates”, *The Journal of Chemical Physics* 161, 124101 (2024). [pdf link](#) (2024 JCP Emerging Investigators Special Collection)

R. Solé, C. P. Kempes, B. Corominas-Murtra, M. De Domenico, **A. Kolchinsky**, M. Lachmann, E. Libby, S. Saavedra, E., D. Wolpert, “Fundamental constraints to the logic of living systems”, *Royal Society Interface Focus* 14, 20240010 (2024). [pdf link](#)

P.M. Riechers, C. Gupta, **A. Kolchinsky**, M. Gu, “Thermodynamically ideal quantum state inputs to any device”, *PRX Quantum* 5, 030318 (2024). [pdf link](#)

A. Kolchinsky, “Partial information decomposition: redundancy as information bottleneck”, *Entropy* 26(7), 546 (2024). [pdf link](#)

J. Piñero, R. Solé, **A. Kolchinsky**[†], “Optimization of nonequilibrium free energy harvesting illustrated on bacteriorhodopsin”, *Physical Review Research* 6, 013275 (2024). [pdf link](#)

A. Kolchinsky[†], N. Ohga, S. Ito, “Thermodynamic bound on spectral perturbations, with applications to oscillations and relaxation dynamics”, *Physical Review Research* 6, 013082 (2024). [pdf link](#)

D.R. Sowinski, J. Carroll-Nellenback, R.N. Markwick, J. Piñero, M. Gleiser, **A. Kolchinsky**, G. Ghoshal, A. Frank, “Semantic Information in a model of resource gathering agents”, *PRX Life* 1, 023003 (2023). [pdf link](#)

A. Kolchinsky[†], “Generalized Zurek’s bound on the cost of an individual classical or quantum computation”, *Physical Review E* 108, 034101 (2023). [pdf link](#)

M. Aguilera and **A. Kolchinsky**, “Quantifying higher-order entropy production in organized nonequilibrium states”, *Proceedings of the ALIFE 2023*, 45 (2023). [pdf link](#)

N. Ohga, S. Ito, **A. Kolchinsky**, “Thermodynamic bound on the asymmetry of cross-correlations”, *Physical Review Letters* 131, 077101 (2023). [pdf link](#) (Editors’ Suggestion; Featured in *Physics*)

K. Yoshimura, **A. Kolchinsky**, A. Dechant, S. Ito, “Housekeeping and excess entropy production for general nonlinear dynamics”, *Physical Review Research* 5, 013017 (2023). [pdf link](#)

F.C. Sheldon, **A. Kolchinsky**, F. Caravelli, “Computational capacity of LRC, memristive, and hybrid reservoirs”, *Physical Review E* 106, 045310 (2022). [pdf link](#)

A. Kolchinsky, “A novel approach to the partial information decomposition”, *Entropy* 24(3), 403 (2022). [pdf code link](#)

A. Kolchinsky[†] and D.H. Wolpert, “Dependence of integrated, instantaneous, and fluctuating entropy production on the initial state in quantum and classical processes”, *Physical Review E* 104, 054107 (2021). [pdf link](#)

A. Kolchinsky[†] and D.H. Wolpert, “Work, entropy production, and thermodynamics of information under protocol constraints”, *Physical Review X* 11, 041024 (2021). [pdf link](#)

A. Kolchinsky[†] and D.H. Wolpert, “Entropy production given constraints on the energy functions”, *Physical Review E* 104, 034129 (2021). [pdf link](#)

A. Kolchinsky and D.H. Wolpert, “Thermodynamic costs of Turing machines”, *Physical Review Research* 2, 033312 (2020). [pdf link](#)

D.H. Wolpert and **A. Kolchinsky**, “The thermodynamics of computing with circuits”, *New Journal of Physics* 22, 063047 (2020). [pdf link](#)

A. Kolchinsky and B. Corominas-Murtra, “Decomposing information into copying versus transformation”, *Royal Society Interface* 17, 20190623 (2020). [pdf link](#)

A.M. Saxe, Y. Bansal, J. Dapello, M. Advani, **A. Kolchinsky**, B.D. Tracey, D.D. Cox, “On the information bottleneck theory of deep learning”, *Journal of Statistical Mechanics*, 124020 (2019). [pdf code link](#)

A. Kolchinsky[†], B.D. Tracey, D.H. Wolpert, “Nonlinear information bottleneck”, *Entropy* 21(12), 1181 (2019). (Entropy 2021 Best Paper Award) [pdf code link](#)

- A. Berdahl, C. Brelsford, C. De Bacco, M. Dumas, V. Ferdinand, J.A. Grochow, L. Hébert-Dufresne, Y. Kallus, C.P. Kempes, **A. Kolchinsky**, D. B. Larremore, E. Libby, E.A. Power, C.A. Stern, B.D. Tracey, “Dynamics of beneficial epidemics”, *Scientific Reports* 9, 15093 (2019). [pdf link](#)
- E.A. Hobson, V. Ferdinand, **A. Kolchinsky**, J. Garland, “Rethinking animal social complexity measures with the help of complex systems concepts”, *Animal Behaviour* 155, 287-296 (2019). [pdf link](#)
- A. Kolchinsky**[†], B.D. Tracey, S. Van Kuyk, “Caveats for information bottleneck in deterministic scenarios”, *ICLR* (2019). [pdf code link](#)
- D.H. Wolpert, **A. Kolchinsky**, J.A. Owen, “A space–time tradeoff for implementing a function with master equation dynamics”, *Nature Communications* 10, 1727 (2019). [pdf link](#)
- A. Avena-Koenigsberger, X. Yan, **A. Kolchinsky**, M. van den Heuvel, P. Hagmann, O. Sporns, “A spectrum of routing strategies for brain networks”, *PLoS Computational Biology* 15(3): e1006833 (2019). [pdf link](#)
- J.A. Owen, **A. Kolchinsky**, D.H. Wolpert, “Number of hidden states needed to physically implement a given conditional distribution”, *New Journal of Physics* 21, 013022 (2019). [pdf link](#) (correction)
- A. Kolchinsky**[†] and D.H. Wolpert, “Semantic information, autonomous agency, and nonequilibrium statistical physics”, *Royal Society Interface Focus* 8, 20180041 (2018). [pdf code link](#)
- A.M. Saxe, Y. Bansal, J. Dapello, M. Advani, **A. Kolchinsky**, B.D. Tracey, D.D. Cox, “On the information bottleneck theory of deep learning”, *ICLR* (2018). [pdf code link](#) (conference version)
- A. Kolchinsky**, N. Dhande, K. Park, Y.Y. Ahn, “The Minor Fall, the Major Lift: Inferring emotional valence of musical chords through lyrics”, *Royal Society Open Science* 4, 170952 (2017). [pdf data code link](#)
- A. Kolchinsky**[†], D.H. Wolpert, “Dependence of dissipation on the initial distribution over states”, *Journal of Statistical Mechanics*, 083202 (2017). [pdf link](#)
- A. Kolchinsky**, B.D. Tracey, “Estimating mixture entropy with pairwise distances”, *Entropy* 19(7), 361 (2017). (correction) [pdf code link](#)
- A. Kolchinsky**[†], A.J. Gates, L.M. Rocha, “Modularity and the spread of perturbations in complex dynamical systems,” *Physical Review E* 92, 060801(R) (2015). [pdf code link](#)
- A. Kolchinsky**, A. Lourenço, H. Wu, L. Li, L.M. Rocha, “Extraction of pharmacokinetic evidence of drug-drug interactions from the literature,” *PLOS One* 10(5): e0122199 (2015). [pdf link](#)
- A. Kolchinsky**, M.P. van den Heuvel, A. Griffo, P. Hagmann, L.M. Rocha, O. Sporns, J. Goñi, “Multi-scale integration and predictability in resting state brain activity,” *Frontiers in Neuroinformatics* 8, 66 (2014). [pdf link](#)
- A. Rossi, F.J. Parada, **A. Kolchinsky**, A. Puce, “Neural correlates of apparent motion perception of impoverished facial stimuli I: A comparison of ERP and ERS activity,” *NeuroImage* 98, 442-459 (2014). [pdf link](#)
- A. Kolchinsky**, A. Lourenço, L. Li, L.M. Rocha, “Evaluation of linear classifiers on articles containing pharmacokinetic evidence of drug-drug interactions,” *Proc Pacific Symposium on Biocomputing*, 409-420 (2013). [pdf link](#)
- A. Kolchinsky** and L.M. Rocha, “Prediction and modularity in dynamical systems,” *Proc of European Conf. on the Synthesis and Simulation of Living Systems (ECAL)*, 65 (2011). [pdf link](#)
- A. Kolchinsky**, A. Abi-Haidar, J. Kaur, A.A. Hamed, L.M. Rocha, “Classification of protein-protein interaction full-text documents using text and citation network features,” *IEEE/ACM Transactions on Computational Biology and Bioinformatics* 7(3), 400-411 (2010). [pdf link](#)

PREPRINTS

- R. Nagayama, **A. Kolchinsky**, S. Ito, “Oscillatory-nonnormal decomposition of dissipation in Ornstein-Uhlenbeck processes”, [arXiv:2606.07263](#) (2026).
- O. M. Miangolarra, R. Sabbagh, **A. Kolchinsky**, “Wasserstein-2 gradient flows and the geometry of entropy production in classical and quantum stochastic thermodynamics”, [arXiv:2606.00698](#) (2026).
- A. Kolchinsky**, N. Ohga, S. Ito, “Cycle affinity and winding localize eigenvalues of Markov generators”, [arXiv:2605.15884](#) (2026).
- J. Korbel, **A. Kolchinsky**, S.A.M. Loos, G. Manzano, R. Garcia-Millan, O. Movilla Miangolarra, É. Roldán, “Quo vadis, stochastic thermodynamics?”, [arXiv:2604.26601](#) (2026).

AWARDS & FUNDING

Fellowships and awards

- 2026 Ramón y Cajal Fellowship (RYC 2025), provisionally selected.
Spanish State Research Agency (AEI)
- 2022 Marie Skłodowska-Curie Individual Fellowship (HORIZON-MSCA-2021-PF-01, #101068029)
“NETOLIFE: Nonequilibrium thermodynamics of the origin of life”
- 2008– Lilly Fellowship in Informatics (competitive doctoral fellowship)
- 2009 Indiana University, funded by the Eli Lilly & Company Foundation

Research funding

- 2023 John Templeton Foundation (#62828), \$947,926, Co-PI
“Goal-directed behavior and the origin of life”
- 2022 John Templeton Foundation (#62417), \$627,227, Co-PI
“Information architectures that enable life: the emergence of meaning”
- 2019 Foundational Questions Institute (FQXi-RFP-IPW-1912), \$118,100, Co-investigator (PI: David H. Wolpert)
“The role of constraints in the thermodynamics of intelligence”
- 2016 Foundational Questions Institute (FQXi-RFP-1622), \$128,319, Co-investigator (PI: David H. Wolpert)
“Observers as self-maintaining non-equilibrium systems”
- 2016 NSF INSPIRE (CHE-1648973), \$999,947, Co-investigator (PI: David H. Wolpert)
“Tradeoffs in the thermodynamics of computation: a new paradigm for biological information-processing”
- 2016 NSF (1620462), \$770,000, Co-investigator (PI: David H. Wolpert)
“Information Networks and the Evolution of Social Organizations”

INVITED TALKS

- 5/2026 *Workshop on Stochastic Thermodynamics VII* (online)
- 3/2026 *Seminar series*, Instituto de Física Interdisciplinar y Sistemas Complejos (IFISC), Mallorca, Spain
- 11/2025 *Kyoto Workshop on Quantum Thermodynamics & Stochastic Thermodynamics*, Kyoto University, Japan
- 10/2025 *Autocatalysis in Reaction Networks Seminar (ARN)* (online)
- 10/2025 *Renormalization for AI Safety Workshop*, Boston, MA, USA
- 9/2025 *ILIAD2: ODYSSEY, conference on mathematical alignment*, Berkeley, CA, USA
- 6/2025 *CRC Colloquium*, Free University of Berlin, Berlin, Germany
- 5/2025 *Joint European Thermodynamics Conference*, Belgrade, Serbia
- 11/2024 *Universal Biology Institute Seminar Series*, University of Tokyo, Japan
- 9/2024 *Information in Matter Colloquium*, AMOLF, Amsterdam, Netherlands
- 7/2024 *Physics Department Seminar*, University of Barcelona, Barcelona, Spain
- 2/2024 *Seminar*, Barcelona Collaboratorium for Modelling and Predictive Biology, Barcelona, Spain
- 11/2023 *Mini-workshop: Non-equilibrium thermodynamics and biology*, Biofisika Institute, Bilbao, Spain
- 11/2023 *Seminar*, Basque Center for Applied Mathematics, Bilbao, Spain
- 7/2023 *Information Engines at the Frontiers of Nanoscale Thermodynamics*, Telluride, CO, USA
- 6/2023 *Webinar: Mathematics, Physics & Machine Learning*, Instituto Superior Técnico, Lisbon, Portugal
- 6/2023 *Conference: Decomposing Multivariate Information in Complex Systems*
Max Planck Institute for the Physics of Complex Systems, Dresden, Germany
- 9/2022 *Seminar*, Dutch Institute of Emergent Phenomena, University of Amsterdam, Netherlands
- 8/2022 *Seminar*, Earth-Life Science Institute (ELSI), Tokyo, Japan
- 6/2022 *Evolution of complexity from the statistical physics perspective*, Yerevan Physics Institute, Armenia
- 6/2022 *Yagami Statistical Physics Seminar*, Keio University, Japan
- 5/2022 *Workshop on Stochastic Thermodynamics III* (online)
- 4/2022 *Cross Labs Workshop: Is AI Extending the Mind?*, Cross Labs, Japan
- 2/2022 *Physics Department Colloquium Series*, University of Rochester, NY, USA
- 12/2021 *Universal Biology Institute Seminar Series*, University of Tokyo, Japan

2/2021 *Workshop: Origins of Life: The Possible and the Actual*, Santa Fe Institute, NM, USA
 7/2020 *ICTP Seminar Series*, Abdus Salam International Center for Theoretical Physics, Trieste, Italy
 5/2020 *Workshop on Stochastic Thermodynamics I* (online)
 2/2020 *AI Seminar Series*, Information Sciences Institute, Los Angeles, CA, USA
 7/2019 *ISTI Seminar Series*, Los Alamos National Lab, Los Alamos, NM, USA
 6/2018 *Connectomics Lecture Series*, Universidad Diego Portales, Santiago, Chile
 4/2018 *Meeting of the Society for the Neural Control of Movement*, Santa Fe, NM, USA
 11/2017 Seoul National University, South Korea
 8/2017 *Workshop: Thermodynamics & Computation: Towards a New Synthesis*, Santa Fe Institute, NM, USA
 10/2016 *Statistical Physics, Information Processing and Biology*, Santa Fe Institute, NM, USA
 2/2016 Information Sciences Institute, Los Angeles, CA, USA

**POPULAR
SCIENCE &
OUTREACH**

Articles

A. Kolchinsky, “The cost of sending a bit across a living cell”, *Physics Magazine*, 2023. [link](#)

Public talks

4/2018 *SITE Santa Fe* (contemporary art museum), Santa Fe, NM, USA
 Public talk on “Life, entropy, and the 2nd law of thermodynamics”

TEACHING

Online courses

2018 “Fundamentals of machine learning” (w/ B.D. Tracey), Santa Fe Institute *Complexity Explorer*
 Designed and delivered a 12-part online course on basics of machine learning ([link](#))

Workshops, lectures, and tutorials

6/2019 Santa Fe Institute Complex Systems Summer School
 Two-part lecture series on machine learning
 3/2019 Santa Fe Institute. Tutorial on “Machine learning with TensorFlow”
 6/2017, Santa Fe Institute. Co-ran tutorial on Python programming and data analysis
 6/2018
 11/2017 Seoul National University, South Korea. Co-organized week-long workshop
 “Thermodynamics, evolution, and inference through the lens of information theory”
 11/2017 ACTION/Trustee Meeting, Santa Fe Institute
 Lecture on “Machine learning: A guide for the perplexed” (w/ B. D. Tracey)
 5/2010 Instituto Gulbenkian de Ciência, Oeiras, Portugal
 Assisted with week-long “Bayesian brain” educational module

Teaching Assistant at Indiana University, Bloomington, IN

Spring 2014 “I400 Large-scale Social Phenomena” with Simon DeDeo ([link](#))
 Spring 2011 “I201 Math and logic foundations of Informatics” with Erik Wennstrom
 Fall 2010 “I485 Biologically Inspired Computing” ([link](#)) with Luis M. Rocha
 Developed and implemented a 5-part problem set for a computational lab
 Fall 2008– “I210 Information Infrastructure” (various instructors)
 Spring 2009 Ran lab to assist with Python programming course

ADVISING

PhD dissertation committees

2025 *André Gomes*, PhD, Electrical & Computer Engineering, Instituto Superior Técnico, Lisbon, Portugal
 Thesis: “Advances in Partial Information Decomposition”

Undergraduate projects supervised

2018 *Nicolas Freitas*, Santa Fe Institute Summer REU Program, Santa Fe, NM, USA
 Project: “Scaling of information in biochemical systems”

2017 *Francis Cavanna*, Santa Fe Institute Summer REU Program, Santa Fe, NM, USA
Project: “Investigating the relationship between criticality and Landauer costs using the Ising model”

**ACADEMIC
SERVICE**

Grant review panels

- European Research Council, Starting and Advanced Grant Panels
- Interdisciplinary Assessment Board for Artificial Intelligence, Spanish Research Agency
- John Templeton Foundation
- Swiss National Science Foundation

Journal refereeing

General: *PNAS*, *Nature Communications*, *npj Complexity*, *Interface Focus*

Physics: *PRL*, *PRX*, *PRX Quantum*, *PRR*, *PRE*, *NJP*, *Frontiers in Physics*, *J of Physics A*, *Physica A*, *EPL*

Biology: *PLoS Comp Bio*, *Proc R Soc B*, *Phil Trans B*, *Theory in the Biosciences*, *BioSystems*

Information theory and machine learning: *NeurIPS*, *ICLR*, *IEEE TPAMI*, *Entropy*, *Kybernetika*, *Applied Sciences*

Guest editor

Entropy special issue on “Thermodynamics and Information Theory of Living Systems” [link](#)

Conference organization

12/2025 Kyoto Workshop on Quantum Thermodynamics and Stochastic Thermodynamics, co-organizer
Kyoto University, Japan ([program](#))

10/2024 Theoretical and Experimental Approaches to Goal-Directed Behavior, co-organizer
Basque Center for Applied Mathematics, Bilbao, Spain ([press](#))

7/2024 Information-Driven States of Matter, co-organizer
University of Rochester, NY, USA ([program](#)) [press](#)

Seminar organization

2023– Co-organized ongoing “Complex Systems Lab” seminar series

2025 [Barcelona Collaboratorium for Modelling and Predictive Biology](#), Barcelona, Spain

2008– Organized a weekly discussion group on complexity and cognitive science ([link](#))

2013 Indiana University, Bloomington, IN, USA

LANGUAGES

Fluent: English, Spanish, Russian.